IILM UNIVERSITY

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Greater Noida, UTTAR PRADESH

ACADEMIC LEARNING REPORT

AICTE VIRTUAL INTERNSHIP PROGRAMME

ARTIFICIAL INTELLIGENCE & MACHINE LEARNING

An Internship Presentation Report on Artificial Intelligence, Machine Learning & Deep Learning

Internship Organization Kodacy

Domain Artificial Intelligence & Machine Learning

Presented By Tanishq
Roll No.:2410031179
Section: 3CSE24

Completed on:- 10th August, 2026

CANDIDATE PROFILE Kodacy • AICTE Virtual Internship

Candidate & Internship Overview

Summary of candidate profile, internship engagement details, and core technical learning tracks.

Candidate & Track Details

Candidate Name: **Tanishq**

Roll No.: **2410031179**

Section: **3CSE24**

Host Organization: **Kodacy**

Domain: **AI & Machine Learning**

Internship Mode: **AICTE Virtual Mode**

Institution: **IILM University, Greater Noida**

Internship Focus Areas

- Artificial Intelligence & Problem-Solving Concepts
- Machine Learning Classifiers & Regression Models
- Deep Learning Fundamentals & Multi-Layer Networks
- Convolutional & Recurrent Neural Architectures (CNN & RNN)
- Python Programming Foundation for Data & AI
- End-to-End Machine Learning Model Development Workflow

ACADEMIC CURRICULUM Kodacy • Structured Learning Program

Internship Profile

Structured learning framework provided by Kodacy for mastering AI & ML concepts.

Structured Learning Experience

The internship provided a comprehensive and structured curriculum in the field of Artificial Intelligence and Machine Learning. The programme focused on developing foundational and conceptual understanding of intelligent systems, statistical decision making, algorithmic models, neural networks, and Python-driven workflows.

Through guided modules, interactive study tracks, and conceptual case reviews, the curriculum bridged theoretical computer science principles with modern data-driven intelligence methods.

Artificial Intelligence
Intelligent Systems

Machine Learning
Supervised & Unsupervised

Deep Learning
Neural Networks

Python Foundation
Programming & Syntax

Modern AI & Applications
Chatbots & Conversational AI Concepts

GOALS Key Competency Pillars

Core Learning Objectives

The primary technical pillars and learning outcomes established for the internship program.

1. AI Fundamentals

Studying the core principles, historical evolution, intelligence classification, and real-world application paradigms of Artificial Intelligence.

2. Machine Learning

Understanding how machines identify patterns from structured data, make probabilistic estimates, and optimize predictive models.

3. Deep Learning

Exploring multi-layered artificial neural networks, activation functions, gradient flow, and specialized CNN & RNN structures.

4. Model Workflow

Learning the foundational steps of data handling, model training, evaluation metrics, and hyperparameter tuning in Python.

Artificial Intelligence Fundamentals

Foundational concepts and primary specialized branches explored within Artificial Intelligence.

Core Concepts

- Introduction to Intelligent Agents
- Key Challenges & Limitations in AI
- Reasoning and Problem-Solving
- Narrow vs. General Intelligence (AGI)

Language & Vision

- Natural Language Processing (NLP)
- Computer Vision Concepts
- Feature Recognition in Images
- Speech & Audio Processing Basics

Cognition & Systems

- Cognitive Computing Paradigms
- Knowledge Representation Schemes
- Expert Systems & Inference Engines
- Ethical & Responsible AI Basics

AI Problem Solving & Intelligent Systems

State space search representations, probabilistic decision frameworks, and algorithmic solutions.

Problem State
Initial problem definition

→

State Space
Representing all states

→

Search Algorithm
Heuristic & tree search

→

Decision Making
Probabilistic evaluation

→

Optimal Solution
Target goal achieved

Search & Game Theory

- State Space Representation & Graph Traversal
- Minimax Algorithm for Two-Player Adversarial Games
- Evolutionary Computation & Genetic Algorithms

Probability & Decision Making

- Probability Fundamentals & Conditional Likelihood
- Bayes' Theorem & Posterior Probability Calculation
- Naive Bayes Classifier for Text & Categorical Data

Machine Learning Concepts

Fundamental learning paradigms, supervised algorithms, and model training concepts.

Raw Data
Input dataset collection

→

Feature Study
Variables & labels

→

Model Training
Fitting algorithms

→

Pattern Learning
Parameter updates

→

Prediction
Output inference

Classification & Regression

- Supervised vs Unsupervised vs Reinforcement Learning
- K-Nearest Neighbours (KNN) Distance-Based Classifier
- Linear Regression for Continuous Target Variables
- Logistic Regression & Sigmoid Curve for Binary Outcomes

Ensemble Methods

- Decision Tree Classification & Information Gain (Entropy/Gini)
- Random Forests: Bagging & Reducing Model Variance
- Gradient Boosting Concepts for Sequential Error Correction
- Introduction to XGBoost for Tabular Data Performance

Deep Learning & Neural Networks

Biological inspirations, multi-layer artificial neurons, and advanced neural architectures.

Artificial Neural Network Anatomy

- **Perceptron Unit:** Inputs, Synaptic Weights, Biases, and Summation
- **Activation Functions:** ReLU, Sigmoid, and Tanh Non-Linearities
- **Multi-Layer Structure:** Input Layer → Hidden Layers → Output Layer
- **Training Concept:** Forward Propagation, Loss Computation & Backpropagation

Specialized Neural Architectures

- **CNN:** Convolution & Pooling for Spatial Feature Extraction
- **RNN:** Recurrent Feedback Loops for Sequential & Temporal Dependencies
- **GANs:** Generator and Discriminator Adversarial Learning Concept

INPUT LAYER
Feature Vectors
→
HIDDEN LAYERS
Weights & Activations
→
OUTPUT LAYER
Probabilities / Classes

CNN (Vision) RNN (Sequence) GAN (Generative)

Convolutional & Recurrent Neural Networks

Understanding architectural differences and typical use cases for CNN and RNN models.

CNN (Convolutional Neural Networks)

Specialized architectures designed for grid-like topology and spatial data such as images.

Image → Feature Extraction → Pooling → Dense Layer → Class

- **Image Classification:** Categorizing visuals into discrete classes
- **Object Detection:** Identifying bounding regions of key objects
- **Computer Vision:** Automated visual understanding
- **Face Recognition:** Feature alignment & facial identification

RNN (Recurrent Neural Networks)

Architectures with internal feedback loops to retain memory of sequential and temporal data.

Sequence → Recurrent Cell → Hidden State → Sequential Output

- **Natural Language Processing:** Sentence token understanding
- **Text Processing:** Sentiment analysis & document indexing
- **Speech Recognition:** Mapping acoustic signals into text
- **Sequential Analysis:** Temporal trend & pattern evaluation

TECHNICAL STACK Module 06 • Programming Foundation

Python Programming for AI & ML

Foundational programming constructs and data structures studied for AI/ML problem solving.

Python Basics & Structures

- **Primitive Data Types:** Integers, Floats, Strings, Booleans
- **Data Collections:** Lists (mutable), Tuples (immutable), Sets (unique)
- **Dictionaries:** Key-value paired storage for structured parameters

Logic & Functionality

- **Control Flow:** Conditional Statements (if/elif/else)
- **Iterative Loops:** for and while loops for data scanning
- **Functions:** Modular functions and anonymous lambda expressions

Python Fundamentals → Data Types → Data Structures → Conditional Statements → Loops & Iterations → Modular Functions → AI & ML Programming

Machine Learning Model Development

Understanding the end-to-end lifecycle and stages involved in building predictive models.

Data Exploration
Inspecting features

→

Preparation
Cleaning & scaling

→

Model Selection
Choosing algorithms

→

Model Building
Fitting training set

→

Validation
Evaluating metrics

→

Fine-Tuning
Adjusting parameters

→

Optimization
Model readiness

Exploration & Prep

Studying statistical distributions, identifying missing entries, and understanding categorical vs numerical feature sets.

Training & Trees

Studying how decision boundaries are formed using Decision Trees and ensemble aggregation in Random Forests.

Validation & Tuning

Learning how to prevent overfitting through cross-validation and hyperparameter tuning of model depth and estimators.

Advanced Machine Learning Concepts

Data pipeline integrity, model validation rigor, and boosting mechanisms.

Data Handling

- Missing value imputation
- One-Hot & Label Encoding
- Scikit-Learn Pipelines

Evaluation Rigor

- K-Fold Cross-Validation
- Precision, Recall & F1-Score
- ROC-AUC Curve Analysis

Gradient Boosting

- Residual learning concept
- Parameter tuning strategies
- XGBoost efficiency concepts

Model Reliability

- Preventing Data Leakage
- Train-Test Contamination
- Generalization testing

Applied & Modern Artificial Intelligence

Conversational systems, foundational Generative AI principles, and real-world implications.

Chatbot Development

- Rule-based vs Intent-driven systems
- Utterance classification & entity parsing
- Dialogue management flowcharts

Understanding ChatGPT

- Concepts of Large Language Models (LLMs)
- Prompt structuring & conversational AI
- Generative text synthesis basics

Present & Future AI

- Applications across healthcare & industry
- Intelligent automation & assistive AI
- Ethical governance & AI safety

Skills & Knowledge Developed

Summary of technical understanding and analytical skills acquired across the internship.

Artificial Intelligence

Understanding intelligent systems, search algorithms, state spaces, and knowledge representations.

Machine Learning

Understanding classification, regression, Bayes rule, Decision Trees, and Random Forests.

Deep Learning

Knowledge of multi-layer perceptrons, activation functions, CNN vision, and RNN sequential models.

Python Foundation

Developing essential programming logic, data structure manipulation, and functional methods.

ML Workflow

Understanding data cleaning, feature exploration, cross-validation, and hyperparameter tuning.

Modern AI & LLMs

Exposure to conversational agents, ChatGPT architecture concepts, and responsible AI practices.

ROADMAP 10-Stage Learning Progression

My AI & ML Learning Journey

Sequential curriculum progression traversed throughout the internship tenure at Kodacy.

STEP 01
AI Fundamentals
History, core branches & intelligent agents

STEP 02
Problem Solving
Search trees, heuristics & state spaces

STEP 03
Probability & Bayes
Statistical decision making & Bayes theorem

STEP 04
Machine Learning
Supervised classification & regression models

STEP 05
Neural Networks
ANN, Perceptrons & backpropagation

STEP 06
CNN & RNN
Computer vision & sequential NLP architectures

STEP 07
Python Foundation
Data structures, loops & function logic

STEP 08
Model Development
Data cleaning, training & validation pipeline

STEP 09
Advanced ML
Gradient Boosting, tuning & data leakage prevention

STEP 10
Modern AI & LLMs
Chatbots, ChatGPT & future AI horizons

Internship Experience & Conclusion

Reflective summary of academic growth, practical insights, and future learning pathways.

Key Internship Reflections

The Artificial Intelligence and Machine Learning virtual internship provided a well-structured and disciplined learning experience covering both theoretical foundations and practical algorithmic workflows.

Throughout the program, I explored intelligent problem solving, statistical decision-making, supervised and unsupervised machine learning algorithms, deep neural architectures, and end-to-end model evaluation techniques in Python.

Academic Value: Established a strong technical foundation for advanced academic coursework and future machine learning projects.

Future Learning Pathways

- **Hands-On Deep Learning Projects**
  Building custom computer vision and NLP models using PyTorch & TensorFlow.
- **Advanced Large Language Models**
  Exploring fine-tuning techniques, prompt engineering, and RAG architectures.
- **MLOps & Deployment Workflows**
  Understanding cloud-based API serving, model tracking, and continuous monitoring.

CREDENTIALS Verification & Records

Formal certificate of completion awarded by Kodacy under the AICTE Virtual Internship Programme.

Credential Details

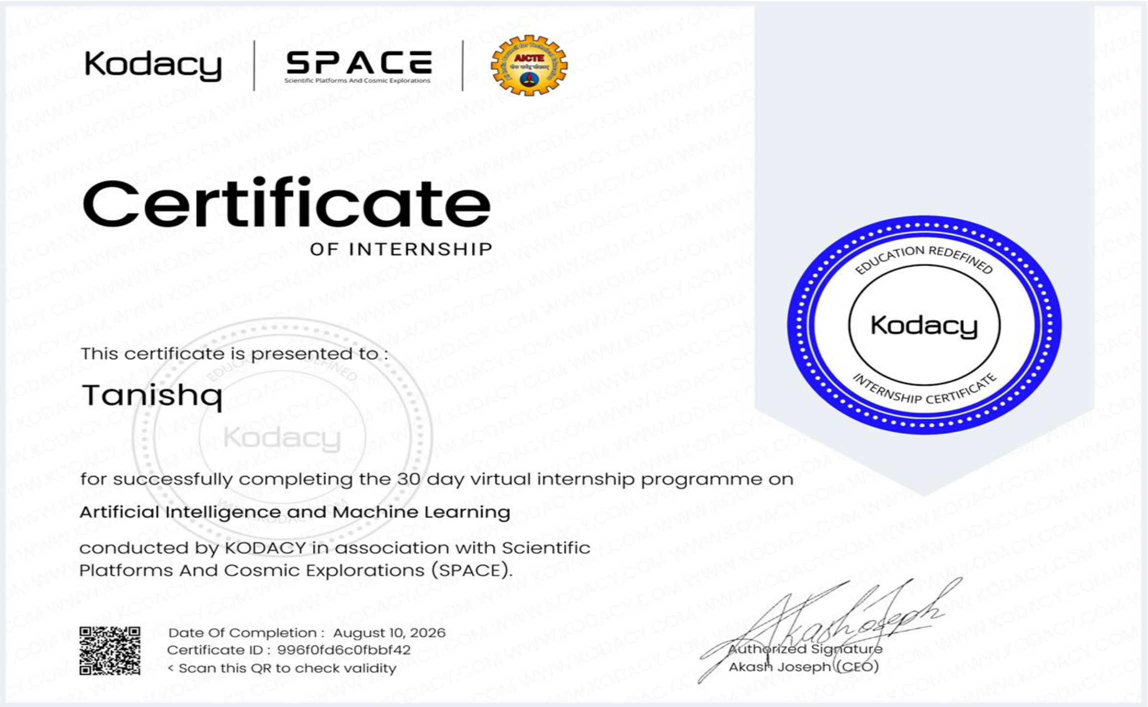
Awarded To: **Tanishq**

Organization: **Kodacy**

Domain: **AI & Machine Learning**

Program: **AICTE Virtual Internship**

Status: **Successfully Completed**



Kodacy | SPACE Scientific Platforms And Cosmic Explorations

Certificate

OF INTERNSHIP

EDUCATION REDEFINED Kodacy INTERNSHIP CERTIFICATE

This certificate is presented to :

Tanishq

for successfully completing the 30 day virtual internship programme on

Artificial Intelligence and Machine Learning

conducted by KODACY in association with Scientific Platforms And Cosmic Explorations (SPACE).

Date Of Completion : August 10, 2026

Certificate ID : 996f0fd6c0fbbf42

< Scan this QR to check validity

Authorized Signature

Akash Joseph (CEO)

Thank You!

Thank you for your time and guidance. Open to questions and academic evaluation.

TANISHQ

Artificial Intelligence & Machine Learning Intern

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